

RISK AVERSION, MEAN REVERSION, AND OPTIMAL TRADING

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Trading activity on the interval $[0, T]$.

price of the asset at time t	:	S_t
the number of shares the trader holds at time t	:	H_t
the wealth of the investor at time t	:	$V_t(H)$
terminal time of trading	:	T

$$V_t(H) = V_{t-1}(H) + H_t (S_t - S_{t-1})$$

model in discrete time

$$\text{Terminal wealth} : V_T(H) = \sum_{t=0}^T H_t (S_t - S_{t-1})$$

model in continuous time

$$\text{Terminal wealth} : V_T(H) = \int_0^T H_u dS_u$$

UTILITY FUNCTION

$$U : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto U(x)$$

- increasing
- concave (maybe with some convex parts)

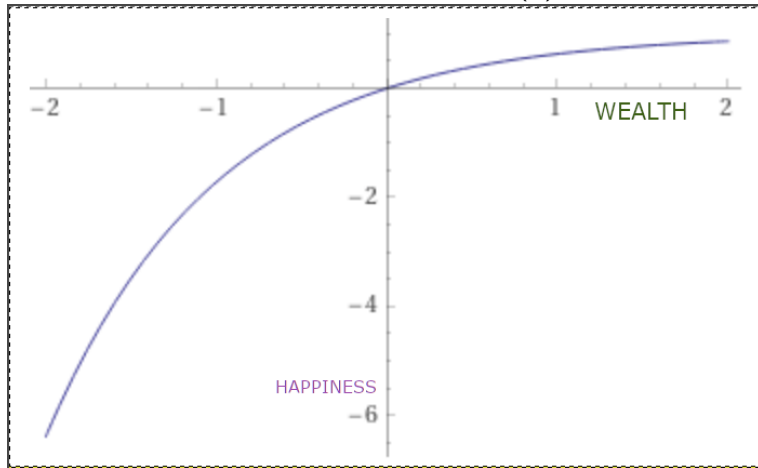
EXAMPLES

isoelastic utility exponential utility

$$U(x) = \frac{x^{1-\eta}-1}{1-\eta} \qquad U(x) = -e^{-x}$$

EXPONENTIAL UTILITY

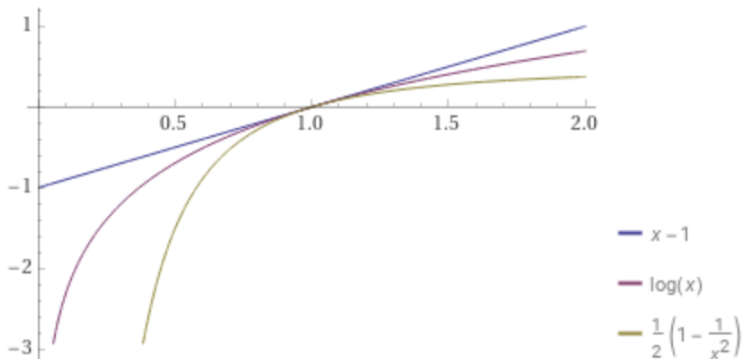
$$U(x) = -e^{-x}$$



$$dU(x) = e^{-x} dx$$

ISOELASTIC UTILITY WITH PARAMETERS $\eta = 0, 1, 3$

$$U(x) = x \quad U(x) = \log(x) \quad U(x) = 1/2 - (1/2)x^{-2}$$



VALUE FUNCTION

$$u(T) = \sup_{H \in \mathcal{X}_T} E \left[U \left(V_T(H) \right) \right]$$

OBJECTIVE

For every T find $H^*(T) \in \mathcal{X}_T$ so that

$$E \left[U \left(V_T(H^*(T)) \right) \right] = u(T)$$

ALTERNATIVELY

Find a family of strategies $\mathfrak{H}^* = (H^*(T))_{T>0}$ so that

$$E \left[U \left(V_T(H^*(T)) \right) \right] \approx u(T)$$

CERTAINTY EQUIVALENT

$$c(T) = U^{-1}\left(u(T)\right) = U^{-1}\left(\sup_{H \in \mathcal{X}_T} E\left[U\left(V_T(H)\right)\right]\right)$$

$$c(T; H) = U^{-1}EU V_T(H)$$

- The investor's opportunity cost of trading.

ORDER OF CERTAINTY EQUIVALENT

For a family of strategies $\mathfrak{H}^* = (H^*(T))_{T>0}$ the order of the certainty equivalent is

$$\mathcal{C}(\mathfrak{H}) = \sup \left\{ \theta : \liminf_{T \rightarrow \infty} \frac{c(T, H(T))}{T^\theta} > 0 \right\}$$

POWER ORNSTEIN-UHLENBECK PROCESS WITH DRIFT

$$dS_t = \mu dt + dX_t,$$

$$dX_t = -\alpha \operatorname{sgn}(X_t) |X_t|^\beta dt + dB_t,$$

$$S_0 = X_0 = 0, \beta \geq 1, \alpha > 0, \mu \in \mathbb{R}, B_t$$

Remark on nonlinear mean reversion

$$d(S_t - \mu t) = -\alpha \operatorname{sgn}(S_t - \mu t) |S_t - \mu t|^\beta dt + dB_t$$

A REFERENCE RESULT

- Merton's 1969 paper presents a strategy that is insensitive to investment horizons

TURNPIKE THEOREMS

- Long horizon optimal investments are time homogeneous
- finite horizon problems converge to their long horizon counterparts
- Merton's result holds for very general utility functions

Guasoni, Paolo and Nagy, Lóránt and Rásonyi, Miklós (2021)
Young, Timid, and Risk Takers. MATHEMATICAL FINANCE

ORNSTEIN-UHLENBECK PROCESS WITH DRIFT, $\beta = 1$

$$dS_t = \mu dt + dX_t,$$

$$dX_t = -\alpha X_t dt + dB_t,$$

Optimal strategy and certainty equivalent

- $H_t = \mu - \alpha X_t + \alpha(\mu - \alpha X_t)(T - t) + \mu\alpha^2(T - t)^2$
- $c(T) = \mu^2 T + (\alpha\mu^2 + \alpha^2)T^2 + \mu^2\alpha^2 T^3$

Performance of Configurations

SETUP	GROWTH OF $c(T)$
$\alpha \neq 0$ and $\mu \neq 0$,	T^3
$\mu = 0$ but $\alpha \neq 0$,	T^2
$\mu \neq 0$ but $\alpha = 0$,	T^1
$\alpha = 0$ and $\mu = 0$	T^0

$$dS_t = \mu dt + dX_t,$$

$$dX_t = -\alpha \operatorname{sgn}(X_t) |X_t|^\beta dt + dB_t,$$

Theorem (NON-STATIONARY REGIME $\mu \neq 0$)

- If $\mu \neq 0$, then

$$c_T \leq C_{\beta, \mu, \alpha} T^{2\beta+1}$$

for some constant $C_{\beta, \mu, \alpha} > 0$, and each family of strategies $\mathfrak{H}$ satisfies

$$\mathcal{C}(\mathfrak{H}) \leq 2\beta + 1.$$

- The family of strategies $\mathfrak{H}_\beta = (H(\beta, T))_{T>0}$ defined by

$$H_t(\beta, T) = (\beta + 1)(T - t)^\beta \operatorname{sgn}(S_t) |S_t|^\beta, \quad t \in [0, T]$$

satisfies

$$\mathcal{C}(\mathfrak{H}_\beta) = 2\beta + 1.$$

$$dS_t = -\alpha \operatorname{sgn}(S_t) |S_t|^\beta dt + dB_t$$

Theorem (STATIONARY REGIME $\mu = 0$)



$$c_T \leq C_{\beta, \alpha} T^{\beta+1}$$

For all $\mathfrak{H}$,

$$\mathcal{C}(\mathfrak{H}) \leq \beta + 1.$$

- If $\beta > 1$, then for each $1 < \gamma < \beta$, the family of strategies $\mathfrak{H}_\gamma = (H(\gamma, T))_{T>0}$ defined by

$$H_t(\gamma, T) = -2(T - t)^\gamma S_t, \quad t \in [0, T]$$

satisfies

$$\mathcal{C}(\mathfrak{H}) \geq 1 + \gamma.$$

- If $\beta = 1$ then

$$\mathcal{C}(\mathfrak{H}_1) = 2 = 1 + \beta.$$